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Vellore Institute of Technology
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School of Electronics Engineering

Quiz -1

BAECE103 – Network Theory

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EXAM DURATION: 30 minutes MAX. MARKS: 10

Q.1.

Q. A load consumes 800 W at a power factor of 0.8 lagging. Its apparent power is:

- A. 640 VA B. 800 VA C. 1000 VAD. 1250 VA

Q.2.

Q. A series RL circuit has $R = 6\Omega$ and $X_L = 8\Omega$. The magnitude of the impedance is:

- A. 6Ω B. 8Ω C. 10Ω D. 14Ω

Q.3.

A voltage $v(t)$ and current $i(t)$ are given by

$$v(t) = 100 \sin \omega t$$

$$i(t) = 5 \sin(\omega t - 30^\circ)$$

The average power consumed by the circuit is:

- A. 250 W B. 216.5 W C. 500 W D. 125 W

Q.4.

Q. If $x(t) = 20 + 15 \cos t + 12 \cos 3t + 10 \cos 5t + \dots$

the magnitude of the dc component is:

- (a) 10, (b) 12, (c) 15, (d) 20

Q.5.

A load has $V = 100\angle 0^\circ \text{ V}$ And $I = 10\angle -30^\circ \text{ A}$.

The complex power consumed by the load is:

- A. $866 + j500\text{VA}$ B. $866 - j500\text{VA}$ C. $500 + j866\text{VA}$
 D. $1000 + j0\text{VA}$

Q.6.

Q. A circuit has $V_{th} = 20\text{V}$ and $Z_{th} = 5\Omega$. If $Z_L = 5\Omega$, the load current is:

- A. 1 A B. 2 A C. 4 A D. 5 A

Q.7.

A single-phase load draws 10A from a 230V supply at a power factor of 0.8 lagging. The real power consumed is:

- A. 1840 W B. 2300 W C. 2875 W D. 1150 W

Q.8.

Q. If the impedance of a circuit is $Z = 10 + j10 \Omega$, the circuit is:

- A. Purely resistive B. Purely capacitive C. Inductive with a 45° impedance angle
 D. Capacitive with a 45° impedance angle

Q.9.

Two AC sources produce currents through a branch:

$$I_1 = 4\angle 0^\circ \text{ A}$$

$$I_2 = 3\angle 90^\circ \text{ A}$$

The magnitude of the resultant current calculated using Superposition Theorem:

- A. 1 A B. 5 A C. 7 A D. 12 A

Q.10.

Q. Maximum Power Transfer Theorem

For maximum average power transfer in an AC circuit, the load impedance Z_L should be:

- A. Equal to Z_{th} B. Equal to Z_{th}^* C. Equal to $-Z_{th}$ D. Equal to $1/Z_{th}$